SECTION 1
ARCHITECTURE AND PASS CREATION

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1.1 PROJECT AIM AND SYSTEM MAP.

Manav built Private Pass, a prototype for offline face checks, to reduce ticket sharing and central face storage (figure 1).

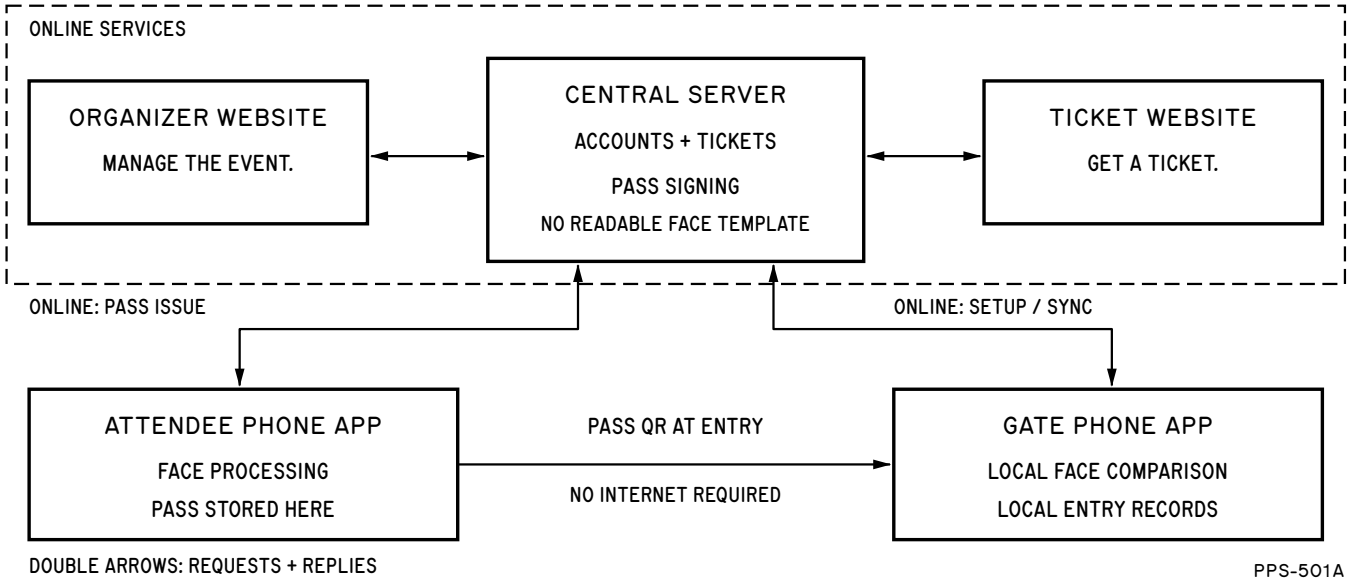


Figure 1. System Architecture: Online Links and Local Entry

PPS

1.2 HOW DOES A FACE BECOME A PASS?

The phone changes a face image into numbers. An event rule makes a template from those numbers. The phone encrypts it with the gate's public key. The server examines ticket ownership and signs the pass. The phone shows the signed pass as a QR code (figure 2).

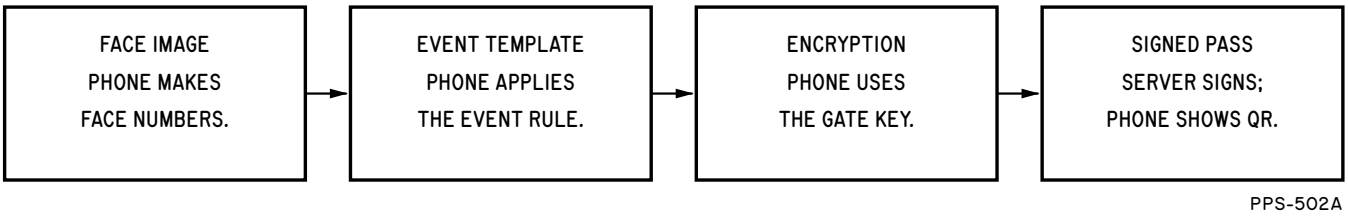


Figure 2. Pass Creation: The Face Stays on the Phone

Table 1. Essential Terms

Term	Meaning
Pass	A digital ticket with encrypted face data.
Template	Numbers used for the face comparison.
Key	Data to lock, open, or verify.
Signature	Data for a source and change check.
Encryption	A change that limits who can read data.
Offline	Without an internet connection.

NOTE

The server receives encrypted data, not the face image or readable template. The public key can be shared. The private key stays on the gate and opens the template. Face data still has privacy risks.

SECTION 2
LOCAL CHECKS AND ENTRY RECORDS

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2.1 HOW DOES THE GATE DECIDE?

The gate first downloads event rules, public keys, and cancellations. It uses these saved items offline. Figure 3 shows the normal acceptance path.

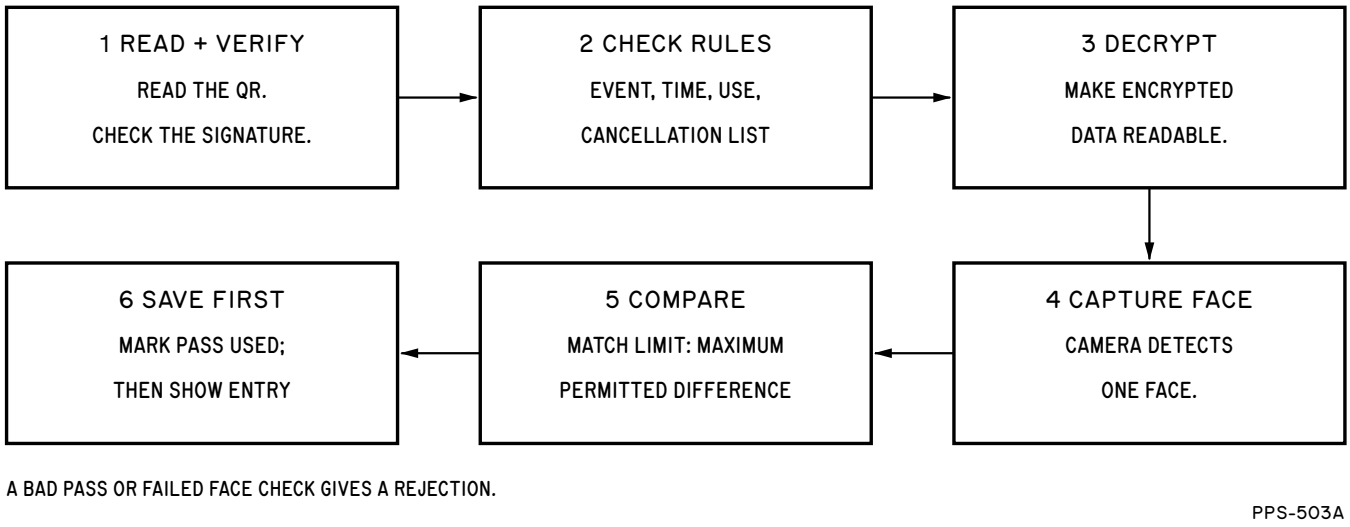


Figure 3. The Gate Checks the Pass, Then the Person

2.2 WHAT HAPPENS INSIDE THE FACE COMPARISON?

The gate applies the same event rule to the new face data. It counts differences between the new template and the pass template. The match limit sets the permitted difference. A small limit can reject the correct person. A large limit can accept the wrong person.

2.3 WHAT HAPPENS TO AN ENTRY RECORD?

The gate saves ticket use and an entry record before it shows entry approval. This stops another use on that gate. The entry record waits for a server connection (figure 4).



Figure 4. Local Storage Keeps the Record Until Reconnection

OFFLINE LIMIT

A gate cannot receive new cancellations while offline. It can use an old cancellation list. A new download is necessary to receive the changes.

WHAT DID I TEST OR ACHIEVE?

A local software test rejected a second entry record for the same pass. This shows database protection against repeat records. It does not test face accuracy or fake-face protection.

Result: gate-db.test.ts, duplicate-record check; verified 28 October 2026. The architecture and normal entry path are simplified.